

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 5-01-210

WASTE DISCHARGE REQUIREMENTS
FOR
NEVADA COUNTY SANITATION DISTRICT NO. 1
PENN VALLEY WASTEWATER TREATMENT PLANT
NEVADA COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. Waste Discharge Requirements Order No. 88-168, adopted by the Board on 23 September 1988, prescribes requirements for the Penn Valley Wastewater Treatment Plant (WWTP), which consists of a wastewater collection, treatment, storage and disposal system (irrigation area). The WWTP property is owned and operated by the Nevada County Sanitation District No. 1 (hereafter Discharger). The Assessor's Parcel Number is 51-310-35.
2. Order No. 88-168 does not reflect current plans and policies of the Board. The purpose of this Order is to rescind Order No. 88-168 and update waste discharge requirements to be consistent with current Board plans and policies.
3. The Penn Valley WWTP is in Sections 4 and 9, T15N, R7E, MDB&M.

Existing Discharge

4. Penn Valley is served by a septic tank effluent pump (STEP) system. Each residence and business has an individual septic tank equipped with a small pump. The septic tanks provide primary treatment. The supernatant from each tank is pumped into a pressurized sewer system to the WWTP. The WWTP consists of two lined aerated lagoons, which may be operated in series or in parallel. Total lagoon volume is approximately 1.27 million gallons (MG). Effluent from the aerated lagoons travels via gravity to a 21-MG lined storage reservoir. The reservoir effluent is used for onsite spray irrigation.
5. The Discharger owns and operates the wastewater collection system infrastructure including the septic tanks. The Discharger periodically inspects, services and pumps the septic tanks on an as needed basis.
6. The design parameters for the Penn Valley WWTP are as follows:

Average Daily Dry Weather Flow:	89,700 gpd
Maximum Daily Dry Weather Flow:	125,600 gpd
Maximum Hour Wet Weather Flow:	228,700 gallons

7. The monthly average peak wet weather flows as reported in the 1999 and 2000 monthly monitoring reports were 107,000 and 92,000 gpd, respectively. Self-monitoring reports from 1998 indicate that winter flows are higher than summer flows, demonstrating significant inflow and infiltration during winter months. Within the next few years, the Discharger plans to propose an ordinance to require lateral pressure testing with needed repairs to be made by the property owner to pass the test.

Hydrology and Soils

8. Previous soil investigations indicate that the site is underlain by highly weathered metavolcanic rock. During an onsite investigation, shallow groundwater seepage was encountered at a depth of approximately 10 feet below grade surface. However, as noted in some of the boreholes drilled for the investigation, the hydraulic gradient in the hillside area caused the groundwater levels to rise in some cases to within a few feet of the ground surface.

Sanitary Sewer System

9. The Discharger's sanitary sewer system collects primary treated wastewater through the use of sewers, pipes, pumps, and/or other conveyance systems and directs this primary treated effluent to the WWTP. A "sanitary sewer overflow" occurs when there is a discharge to ground or surface water from the sanitary sewer system at any point upstream of the wastewater treatment plant. Temporary storage and conveyance facilities (such as wet wells, regulated impoundments, tanks, highlines, etc.) are part of the sanitary sewer system and discharges to these facilities are not considered sanitary sewer overflows, provided that the waste is fully contained within these temporary storage/conveyance facilities.
10. Sanitary sewer overflows may consist of varying mixtures of domestic sewage, industrial wastewater, and commercial wastewater; this mixture depends on pattern of land use in the sewage collection system tributary to the overflow. The chief causes of sanitary sewer overflows include grease blockages, root blockages, debris blockages, sewer line flood damage, manhole structure failures, vandalism, pump station mechanical failures, power outages, storm or groundwater inflow/infiltration, lack of capacity, and contractor caused blockages.
11. Sanitary sewer overflows may contain high levels of suspended solids, pathogenic organisms, toxic pollutants, nutrients, oxygen demanding organic compounds, oil and grease, and other pollutants. Sanitary sewer overflows can cause temporary exceedances of applicable water quality objectives, pose a threat to public health, adversely affect aquatic life, and impair the public recreational use and aesthetic enjoyment of surface waters in the area.
12. The Discharger is expected to take all necessary steps to adequately maintain, operate, and prevent discharges from its sanitary sewer collection system. This Order requires the Discharger to prepare and implement a sanitary sewer operation and maintenance plan, a sanitary sewer overflow prevention plan and a sanitary sewer overflow response plan.

Groundwater Monitoring Data

13. There are four monitoring wells installed at the facility. Monthly sampling results submitted to the Regional Board pursuant to Monitoring and Reporting Program Order No. 88-168 for 1998, 1999 and 2000, indicate the following averages:

Constituent	MW-1 (range)	MW-2 (range)	MW-3 (range)	MW-4 (range)
Nitrogen (mg/L)	0.5 (<0.1 to 0.9)	0.5 (<0.1 to 0.9)	0.4 (<0.1 to 0.9)	0.7 (<0.1 to 2.4)
TDS (mg/L)	136 (53 to 173)	153 (75 to 253)	164 (133 to 226)	161 (106 to 200)
EC (µmhos/cm)	210 (110 to 264)	255 (170 to 480)	238 (203 to 282)	243 (190 to 290)

Degradation

14. State Water Resources Control Board (SWRCB) Resolution No. 68-16 (hereafter Resolution 68-16 or the “Antidegradation” Policy) requires the Board in regulating the discharge of waste to maintain high quality waters of the State until it is demonstrated that any change in quality will be consistent with maximum benefit of the people of the State, will not unreasonably affect beneficial uses and will not result in water quality less than that is described in the Board’s policies (e.g. quality that exceeds water quality objectives).
15. The Board finds that some degradation of groundwater beneath the WWTP and irrigation area is consistent with Resolution 68-16 provided that:
- The degradation is confined to a specified area;
 - The Discharger minimizes the degradation by fully implementing, regularly maintaining, and optimally operating best practicable treatment and control (BPTC) measures;
 - The degradation is limited to waste constituents typically encountered in municipal wastewater as specified in the groundwater limitations in this Order; and
 - The degradation does not result in water quality less than that prescribed in the Basin Plan.
16. Some degradation of groundwater by some of the typical waste constituents released with discharge from a municipal wastewater utility after effective source control, treatment, and control is consistent with maximum benefit to the people of California. The technology, energy, water recycling, and waste management advantages of municipal utility service far exceed any benefits derived from a community otherwise reliant on numerous concentrated individual wastewater systems, and the impact on water quality will be substantially less. Degradation of groundwater by constituents (e.g., toxic chemicals) other than those specified in the groundwater limitations in this Order, and by constituents that can be effectively removed by conventional

treatment (e.g., BOD, total coliform bacteria) is prohibited. When allowed, the degree of degradation allowed depends upon many factors (e.g., background water quality, the waste constituent, the beneficial uses and most stringent water quality objective, source control measures and waste constituent treatability).

17. Economic prosperity of local communities and associated industry is of maximum benefit to the people of California, and therefore sufficient reasons exists to accommodate growth and groundwater degradation around the WWTP, provided that the terms of the Basin Plan are met.

Treatment and Control Practice

18. These waste discharge requirements do not allow degradation of groundwater beneath the irrigation area. These requirements require the Discharger to monitor the groundwater at the irrigation area and if the monitoring data indicate that the discharge of waste to the irrigation area is causing groundwater to contain waste constituents in concentrations statistically greater than background water quality, then the Discharger may be required to submit a report to indicate how such degradation will comply with Resolution No. 68-16. Upon review of such report, the Board may revise this Order, including the groundwater limitation section.
19. The Penn Valley WWTP as described in Finding No. 4, provides the following BPTCs:
 - a). Technology for secondary treatment of municipal wastewater;
 - b). Consistent solids removal prior to treatment;
 - c). Lined, aerated treatment lagoons and reservoir; and
 - d). Operations and Maintenance Manual.
20. The treatment of the WWTP is being operated within the treatment requirements as outlined in Order No. 88-168. The mean of the 12-month average treated effluent BOD concentrations for the year 2000 was 14.6 mg/L. The highest concentration for the year 2000 was 24.5 mg/L.
21. The following best management practices (BMPs) are performed district-wide by the Discharger:
 - a). Periodic inspection and cleaning of the septic tanks and individual septic tank pumps;
 - b). Scheduled inspection and cleaning of pump stations;
 - c). Scheduled exercise and maintenance of emergency generators; and
 - d). Inspection of each individual STEP system, including tank, pump controls and appurtenances.
22. This Order establishes schedules of tasks to evaluate BPTC for each treatment, storage and disposal component of the WWTP and to characterize groundwater for all constituents.
23. This Order establishes interim groundwater limitations for the WWTP that will not unreasonably threaten present and anticipated beneficial uses or result in groundwater quality that exceeds

water quality objectives set forth in the Basin Plan. This Order contains tasks for assuring that BPTC and the highest water quality consistent with the maximum benefit to the people of the State will be achieved. Accordingly, the discharge is consistent with the antidegradation provisions of Resolution 68-16. Based on the results of the scheduled tasks, the Board may reopen this Order to reconsider groundwater limitations and other requirements to comply with Resolution 68-16.

Groundwater Considerations

24. California Department of Water Resources standards for the construction and destruction of groundwater wells (hereafter DWR Well Standards), as described in *California Well Standards Bulletin 74-90* (June 1991) and *Water Well Standards: State of California Bulletin 94-81* (December 1981), and any more stringent standards adopted by the Discharger or county pursuant to California Water Code (CWC) Section 13801, apply to all monitoring wells.
25. State regulations that prescribe procedures for detecting and characterizing the impact of waste constituents from waste management units on groundwater are found in Title 27, California Code of Regulations (CCRs) (hereafter Title 27). While the wastewater treatment plant is exempt from Title 27, the data analysis methods of Title 27 are appropriate for determining whether the discharge complies with the terms for protection of groundwater specified in this Order.

Basin Plan Considerations

26. *The Water Quality Control Plan for the Sacramento River Basin and San Joaquin River Basin, Fourth Edition*, (hereafter Basin Plan), designates beneficial uses, establishes water quality objectives, contains implementation plans and policies for protecting waters of the basin, and incorporates by references plans and policies adopted by the State Water Resources Control Board. These requirements implement the Basin Plan.
27. The Basin Plan encourages reclamation of wastewater wherever feasible.
28. The Basin Plan establishes numerical and narrative water quality objectives for surface and groundwaters within the basin, and recognizes that water quality objectives are achieved primarily through the Board's adoption of waste discharge requirements and enforcement orders. Where numerical water quality objectives are listed, these are limits necessary for the reasonable protection of beneficial uses of the water. Where compliance with narrative water quality objectives is required, the Board will, on a case-by case basis, adopt numerical limitations in orders which will implement the narrative objectives to protect beneficial uses of the waters of the state.
29. The Basin Plan identifies numerical water quality objectives for waters designated as municipal supply. These are the maximum contaminant levels (MCLs) specified in the following provisions of Title 22, California Code of Regulations: Tables 64431-A (Inorganic Chemicals)

and 64431-B (Fluoride) of Section 64431, Table 64444-A (Organic Chemicals) of Section 64444, and Table 64449-A (Secondary Maximum Contaminant Levels-Consumer Acceptance Limits) of Section 64449. The Basin Plan's incorporation of these provisions by reference is prospective, and includes future changes to the incorporated provisions as the changes take effect. The Basin Plan recognizes that the Board may apply limits more stringent than MCLs to ensure that waters do not contain chemical constituents in concentrations that adversely affect beneficial uses.

30. The Basin Plan contains narrative water quality objectives for chemical constituents and toxicity. The toxicity objective requires that groundwater be maintained free of toxic substances in concentrations that produce detrimental physiological responses in plants or animals. The chemical constituent objective requires that groundwater shall not contain chemical constituents in concentrations that adversely affect beneficial uses.
31. Section 13241 of the Water Code requires the Regional Board to consider various factors, including economic considerations, when adopting water quality objectives into its Basin Plan. Water Code Section 13263 requires the Regional Board to address the factors in Section 13241 in adopting waste discharge requirements. The State Board, however, has held that a Regional Board need not specifically address the Section 13241 factors when implementing existing water quality objectives in waste discharge requirements because the factors were already considered in adopting water quality objectives. These waste discharge requirements implement adopted water quality objectives. Therefore, no additional analysis of the Section 13241 factors is required.

Beneficial Uses

32. Surface water drainage is to Tarr Ditch, tributary to Little Dry Creek, thence the Bear River.
33. The beneficial uses of the Bear River are municipal, industrial, and agricultural supply; recreation; aesthetic enjoyment; ground water recharge; fresh water habitat, hydroelectric power generation; and preservation and enhancement of fish, wildlife and other aquatic resources.
34. The beneficial uses of ground waters are municipal, domestic, industrial, and agricultural supply.

Sludge and Solid Waste

35. The discharge authorized herein and the treatment and storage facilities associated with the discharge, except for discharges of residual sludge and solid waste, are exempt from the requirements of Title 27, CCR, Section 20380 et seq. The exemption, pursuant to Title 27, CCR, Section 20090(a), is based on the following:
 - a). The waste consists primarily of domestic sewage and treated effluent;
 - b). The waste discharge requirements are consistent with water quality objectives; and

- c). The treatment and storage facilities described herein are associated with a municipal wastewater treatment plant.

Water Recycling

36. The California Department of Health Services (DHS) has established statewide criteria in Title 22, CCR, Section 60301 et. seq. (hereafter Title 22). DHS revised the water recycling criteria contained in Title 22 on 2 December 2000. The revised Title 22 required that all wastewater used for reclamation receive, at a minimum, secondary treatment. However, Title 22 does not define secondary treatment with respect to numerical limits for BOD₅ and total suspended solids (TSS). According to DHS, for uses not requiring disinfection, treated wastewater should, at a minimum, be adequately oxidized, contain dissolved oxygen near saturation levels, and be nonputrescible when applied to land.
37. Title 22, Section 60303 states that water recycling requirements shall not apply to the use of recycled water onsite at a water recycling plant, or wastewater treatment plant, provided access by the public to the area of onsite recycled water use is restricted.
38. Title 22, Section 60323(a) states that no person shall produce or supply reclaimed water for direct reuse from a proposed water reclamation plant unless an engineering report is submitted for review and approval by DHS and the Board. Fodder and fiber crops and pasture for animals not producing milk for human consumption is considered a beneficial reuse. A Title 22 Engineering Report shall be submitted to the Board and DHS for review and approval prior to grazing livestock in the irrigation pasture land. Provision E.1.c outlines a compliance schedule for the submittal of the Title 22 Engineering Report.
39. DHS has recommended that storm water discharges from wastewater recycling areas be fully contained and returned year round to storage ponds. Storm water runoff from these areas can be allowed provided the following conditions:
- a. Disinfection of the wastewater to a 23 MPN/100 ml monthly median with a 240 MPN/100 ml daily maximum is to be maintained or disinfection of the irrigation area prior to storm water release as approved by DHS.
 - b. Application of disinfected wastewater is ceased a minimum of 24 hours prior to a diversion of storm water to surface water drainage courses.
 - c. During wastewater application, all tailwater and/or storm water shall be returned to the wastewater ponds.
 - d. All wastewater is absorbed into the soil of the irrigation area, tailwater collection and return system, or returned to the wastewater ponds within 24 hours after application has ceased.
 - e. No standing water is allowed at any time in the tailwater collection and return system.
 - f. Any commingled undisinfected wastewater and storm water must be contained and treated.

Other Regulatory Considerations

40. Section 13267(b) of California Water Code provides that: “In conducting an investigation specified in subdivision (a), the regional board may require that any person who has discharged, discharges, or is suspected of discharging, or who proposes to discharge within its region, or any citizen or domiciliary, or political agency or entity of this state who has discharged, discharges, or is suspected of discharging, or who proposes to discharge waste outside of its region that could affect the quality of the waters of the state within its region shall furnish, under penalty of perjury, technical or monitoring program reports which the board requires. The burden, including costs of these reports, shall bear a reasonable relationship to the need for the reports and the benefits to be obtained from the reports.”
41. Pursuant to CWC Section 13263 (g), discharge is a privilege, not a right, and adoption of this Order does not create a vested right to continue the discharge.
42. The action to update waste discharge requirements for this facility is exempt from the provisions of the California Environmental Quality Act (CEQA), in accordance with the CCR, Title 14, Section 15261. This exemption reflects the ongoing nature of the wastewater treatment plant and does not provide for any net increase in treatment or disposal capacity from Order No. 88-168.

Public Notice

43. The Board considered all the above and the supplemental information and details in the attached Information Sheet, which is incorporated by reference herein, in establishing the following conditions of discharge.
44. The Board consulted with the Department of Health Services, and considered their recommendations regarding public health aspects for water recycling on the irrigation area.
45. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge.
46. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 88-168 is rescinded and that, pursuant to CWC sections 13263 and 13267, Nevada County Sanitation District No. 1, its agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the CWC and regulations adopted there under, shall comply with the following at the Penn Valley Wastewater Treatment Plant.

[Note: Other prohibitions, conditions, definitions, and some methods of determining compliance are contained in the “Standard Provisions and Reporting Requirements for Waste Discharge Requirements” dated 1 March 1991, which is attached hereto and made part of this Order by reference.]

A. Discharge Prohibitions:

1. The direct discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. The discharge of effluent to the irrigation area during a storm event is prohibited.
3. The by-pass or overflow of untreated or partially treated waste is prohibited.
4. Public contact with wastewater effluent is prohibited.
5. Cattle grazing in the irrigation area will be prohibited after **1 January 2003** if a Title 22 Engineering report has not been reviewed and approved by DHS and the Executive Officer. Provision E.1.c outlines the compliance schedule for the submittal of the report.
6. The acceptance of septage is prohibited.
7. Discharge of waste classified as hazardous, as defined in Sections 2521(a) of Title 23, CCR, Section 2510, et seq., (hereafter Chapter 15), or 'designated', as defined in Section 13173 of the CWC, is prohibited.

B. Discharge Specifications:

1. Neither the treatment nor the discharge shall cause a pollution or nuisance as defined by the Porter-Cologne Water Quality Control Act, Section 13050.
2. The discharge shall not cause degradation of any water supply.
3. The following design flow limitations shall not be exceeded:

Average Daily Dry Weather Flow:	89,700 gpd
Maximum Daily Dry Weather Flow:	125,600 gpd
Maximum Hour Wet Weather Flow:	228,700 gallons
4. During the dry season all wastewater and storm water tailwater shall be returned to the storage reservoir.
5. The maximum total nitrogen loading to the irrigation area shall not exceed the agronomic rate for plant available nitrogen (PAN) for the type of crop to be grown, as specified in the most recent edition of the Western Fertilizer Handbook.
6. Wastewater shall be applied to the irrigation area such that there will be no standing water in the irrigation area or in the wastewater runoff (tailwater) collection and return system

- 24 hours after wastewater application has ceased. Wastewater application to the irrigation area shall be managed to minimize erosion and runoff.
7. Wastewater shall not migrate from the WWTP or irrigation area.
 8. The irrigation area shall be managed to prevent breeding of mosquitoes.
 9. All tailwater shall be collected and returned to the storage reservoir when wastewater is being applied to the irrigation area. Wastewater application shall cease 48 hours prior to an anticipated storm event and may resume 48 hours after a storm event. Wastewater application may resume at any time when weather conditions permit and full tailwater control is in effect.
 10. The Discharger shall cease application of wastewater when winds exceed 30 mph.
 11. The Discharger shall comply with applicable regulation criteria set forth in Title 22, Division 4, Section 60301 et seq., CCR.
 12. Public contact with wastewater shall be precluded through such means as fences, placards, and/or irrigation management practices.
 13. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater treatment and disposal areas.
 14. As a means of discerning compliance with Discharge Specification B.13, the dissolved oxygen content in the upper zone (1 foot) of the aeration lagoons and storage reservoir shall not be less than 1.0 mg/l.
 15. Recycled water applied to the irrigation area shall be disinfected or the irrigation area shall be disinfected to protect human health and surface waters from the release of storm water from the irrigation area. The Disinfection method shall be reviewed and approved by DHS and the Executive Officer. Provision E.1.b outlines a compliance schedule for the implementation of the disinfection method(s).
 16. The effluent discharged to the irrigation area shall not have a pH less than 6.5 or greater than 9.0.
 17. The following constituent limitations shall apply to wastewater discharged to the irrigation area:

<u>Constituent</u>	<u>Units</u>	<u>30-Day Average</u>	<u>30-Day Median</u>	<u>Daily Maximum</u>
BOD ¹	mg/l	40	---	80
Settleable Solids	ml/l/hr	0.5	---	1.0
Total Coliform Organisms ²	MPN ³ /100 ml	---	23	240

¹ 5-day biochemical oxygen demand at 20° C.

² Coliform analysis shall be performed upon implementation of the disinfection alternative as required by Provision E.1.b of this Order.

³ Most Probable Number

18. The aerated lagoons and storage reservoir shall be managed to prevent breeding of mosquitoes. In particular:
 - a. An erosion control program should assure that small coves and irregularities are not created around the perimeter of the water surface.
 - b. Weeds shall be minimized through control of water depth, harvesting, or herbicides.
 - c. Dead algae, vegetation, and debris shall not accumulate on the water surface.
19. A controlled 100-foot buffer shall be maintained around the irrigation area, and between any watercourse and the wetted area produced during spray disposal.
20. Signs with proper wording of sufficient size shall be placed at areas of access and around the perimeter of all areas used for effluent disposal to alert the public of the presence of wastewater.
21. WWTP and irrigation area design shall be constructed, operated, and maintained to prevent inundation or washout due to flood, storm, or an entire winter season with a 100-year return frequency.
22. The WWTP shall have sufficient capacity to accommodate allowable wastewater flow and design seasonal precipitation and ancillary inflow and infiltration during the non-irrigation season and collection of all tailwater and storm water from the irrigation area to comply with Discharge Specification B.21. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100-years, distributed monthly in accordance with historical rainfall patterns.
23. The aerated lagoons and storage reservoir freeboard shall never be less than two feet (measured vertically to the lowest point of overflow).
24. Before **1 October** of each year, available WWTP storage capacity shall at least equal the volume necessary to comply with Discharge Specifications Nos. 22 and 23.

25. Upon the reduction, loss, or failure of the sanitary sewer system resulting in a sanitary sewer overflow, the Discharger shall, to the extent necessary to maintain compliance with this Order, take any necessary remedial action to (a) control or limit the volume of sewage discharged, (b) terminate the sewage discharge as rapidly as possible, and (c) recover as much as possible of the sewage discharged (including washdown water) for proper disposal. The Discharger shall implement all remedial actions to the extent they may be applicable to the discharge including the following:
 - a. Interception and rerouting of sewage flows around the sewage line failure;
 - b. Vacuum truck recovery of sanitary sewer overflows and washdown water;
 - c. Use of portable aerators where complete recovery of the sanitary sewer overflows are not practicable and where severe oxygen depletion is expected in surface waters; and
 - d. Cleanup of sewage-related debris at the overflow site.
26. No waste constituent shall be released or discharged, or placed where it will be released or discharged, in a concentration or in a mass that causes violation of Groundwater Limitations.

C. Sludge Specifications:

1. Sludge in this document means the solid, semisolid, and liquid residues removed during primary, secondary, or advanced wastewater treatment processes. Solid waste refers to grit and screening material generated during preliminary treatment. Residual sludge means sludge that will not be subject to further treatment at the WWTP. Biosolids refers to sludge that has undergone sufficient treatment and testing to qualify for reuse pursuant to federal and state regulations as a soil amendment for agriculture, silviculture, horticulture and land reclamation.
2. Sludge and solid waste shall be removed from screens, sumps, septic tanks, ponds as needed to ensure optimal WWTP operation and adequate hydraulic capacity.
3. Treatment and storage of sludge generated by the WWTP shall be confined to the WWTP property and conducted in a manner that precludes infiltration of waste constituents into soils in a mass or concentration that will violate Groundwater Limitations.
4. Any storage of residual sludge, solid waste and biosolids on property of the WWTP shall be temporary and controlled and contained in a manner that minimizes leachate formation and precludes infiltration of waste constituents into soils in a mass or concentration that will violate Groundwater Limitations.
5. Residual sludge, biosolids and solid waste shall be disposed of in a manner approved by the Executive Officer and consistent with Title 27. Removal for further treatment,

disposal, or reuse at sites (e.g., landfill, WWTP, composting site, soil amendment sites) operated in accordance with valid waste discharge requirements issued by a Regional Water Quality Control Board will satisfy this specification.

6. Use of biosolids as a soil amendment shall comply with General Biosolids Order (State Water Resources Control Board Water Quality Order No. 2000-10-DWQ, General Waste Discharge Requirements for the Discharge of Biosolids to Land as a Soil Amendment in Agriculture, Silviculture, Horticultural, and Land Reclamation Activities). The Discharger must obtain a Notice of Applicability of the General Biosolids Order from the Executive Officer or site-specific waste discharge requirements prior to discharge of biosolids to any site.
7. Use and disposal of biosolids should comply with the self-implementing federal regulations of 40 CFR 503, which are subject to enforcement by the U.S. Environmental Protection Agency (EPA), not the Board. If during the life of this Order the State accepts primacy for implementation of 40 CFR 503, the Board may also initiate enforcement where appropriate.

D. Groundwater Limitations:

1. Release of waste constituents from any wastewater treatment or storage system component associated with the WWTF shall not cause groundwater under and beyond that system component, as determined by an approved well monitoring network, to:
 - a. Contain any of the following constituents in concentration greater than as listed or greater than background quality, whichever is greater:

<u>Constituent</u>	<u>Units</u>	<u>Limitation</u>
Boron	mg/L	0.6
Chloride	mg/L	106
Iron	mg/L	0.3
Manganese	mg/L	0.05
Total Coliform Organisms	MPN/100 ml	Nondetect
Total Dissolved Solids ¹	mg/L	450
Total Nitrogen	mg/L	10
Nitrite (as N)	mg/L	1
Nitrate (as N)	mg/L	10
Ammonia (as N)	mg/L	0.5
Total Organic Carbon (TOC)	mg/L	0
Total Trihalomethanes	µg/L	100

¹ A cumulative impact that accounts for several dissolved constituents in addition to those listed here separately [e.g. alkalinity (carbonate and bicarbonate), calcium, hardness, phosphate, potassium, etc.].

- b. Contain any constituent not identified in Groundwater Limitation D.1.a in concentrations greater than background quality (whether chemical, physical, biological, bacteriological, radiological, or some other property or characteristic).
 - c. Exhibit a pH of less than 6.5 or greater than 8.5 pH units.
 - d. Impart taste, odor, or color that creates nuisance or impairs any beneficial use.
2. For the effluent recycling and irrigation application areas:
- a. Release of waste constituents from any irrigation application area associated with the WWTP shall not cause groundwater under and beyond the irrigation area to contain waste constituents in concentration statistically greater than background water quality.
 - b. If groundwater monitoring shows that waste constituents are present in concentrations greater than background, then upon request of the Executive Officer, the Discharger shall complete the report described in Provision E.3.

E. Provisions:

1. All of the reports shall be submitted pursuant to Section 13267 of the CWC. All reports shall be prepared and stamped by a California Registered Engineer or Engineering Geologist:
- a. Submit by **1 October 2001** a scaled map of the wastewater treatment facility including the locations of the on-site groundwater monitoring wells. Include within this submittal the monitoring well logs as well as the professionally surveyed monitoring well elevations.
 - b. Submit by **1 January 2002** a work plan, which outlines the method by which the Discharger will disinfect the wastewater effluent or irrigation application area to achieve the effluent limits on total coliform of 23 MPN/100 ml monthly median and 240 MPN/100 ml daily maximum or demonstrate that that the proposed disinfection method(s) will protect human health and water quality protection in the event of storm water run off. The disinfection work plan shall include the disinfection schemes during the dry and wet seasons and implementation schedule. This plan shall be reviewed and approved by DHS. The approved disinfection alternative(s) shall be implemented by **15 March 2003**.
 - c. If it is the Discharger's intention to continue to graze cattle in the irrigation area, then a Title 22 Engineering Report shall be submitted to DHS and the Board for review and approval by **1 August 2002**. Cattle may be grazed until this compliance date. If the report is not submitted and approved by **1 January 2003**, then in accordance with

Title 22 the Discharger will be prohibited from grazing cattle in the irrigation disposal area.

- d. Submit by **1 January 2003** an evaluation, prepared by a registered engineer or geologist, of the existing monitoring well network. The monitoring well network shall consist of one or more background monitoring wells and compliance monitoring wells downgradient of the treatment, storage and irrigation area. If the existing monitoring well network is not sufficient to adequately monitor groundwater conditions (background and downgradient water quality), then additional monitoring wells may need to be installed and a monitoring well network installation work plan that satisfies the second section of Attachment B shall be submitted to be approved by the Executive Officer.
- e. If additional groundwater monitoring wells are warranted then the following compliance schedule shall be implemented following the approval of the work plan:

<u>Task</u>	<u>Compliance Date</u>
i. Implement Monitoring Well Installation Plan	150 days following work plan approval by Executive Officer
ii. Complete Monitoring Well Installation	60 days following Work Plan implementation
iii. Submit Monitoring Well Installation Report of Results	30 days following Project Completion
iv. Commence Groundwater Monitoring	30 days following Project Completion

- f. After satisfying Provisions E.1.d and E.1.e, the Discharger shall commence monitoring in accordance with the groundwater monitoring program described in the MRP for one full year at least at the frequency specified in the MRP. Within 90 days of completing one full year of groundwater sampling, the Discharger shall submit a written technical report that characterizes the groundwater quality of each monitoring well. The technical report shall be prepared and certified by a California registered civil engineer or registered geologist. The report shall indicate for each constituent identified in the MRP, the background concentration in the background well(s) and the actual concentration in each compliance monitoring well. Determination of background water quality shall be made using the methods described in Title 27, Section 20415(e)(10). The report shall compare actual concentrations of Groundwater Limitations D.1 and D.2 and report the compliance results. For purposes of the Report, the Discharger will recommend background limitations for waste constituents not listed Groundwater Limitations D.1 and for those listed where background concentrations are greater than identified. Subsequent use of a concentration as a background limitation will be subject to the discretion of the Executive Officer.

Antidegradation Analysis Evaluation

- g. By **1 August 2003**, the Discharger shall submit for Executive Officer approval a written work plan in the form of a technical report that sets forth a schedule for systematic and comprehensive technical evaluation of each component of the WWTP's waste treatment and control to determine for each waste constituent BPTC as used in Resolution 68-16. The technical report shall contain a preliminary evaluation of each component and propose a time schedule for completing the comprehensive technical evaluation. The technical report shall be prepared and certified by a California registered civil engineer. The schedule to complete all comprehensive technical evaluations shall be as short as practicable and shall not exceed one year.
- h. By the schedule approved by the Executive Officer pursuant to Provision E.1.g, but no later than **1 March 2005**, the written comprehensive technical evaluation shall be submitted with the Discharger's written recommendations for the WWTP modifications (e.g. component upgrade and retrofit). The report shall include specific methods the Discharger proposes as a means to measure processes and assure continuous optimal performance of BPTC measures. Comprehensive technical evaluations shall be prepared and certified by a California registered civil engineer. The source of funding and proposed schedule for modifications shall be identified. The schedule shall be as short as practicable but in no case shall exceed five years past the Executive Officer's approval of the comprehensive technical evaluation unless the schedule is reviewed and specifically approved by the Board. The component evaluation, recommended improvements and schedule are subject to the Executive Officer's review and approval.
- i. By **1 August 2005**, the Discharger shall submit a technical report that proposes specific numeric groundwater limitations that reflect full implementation of BPTC and describe how these were determined considering actual data from compliance monitoring wells, impact reductions through full implementation of BPTC, reasonable growth, etc. The Discharger should submit results of a validated groundwater model to support its proposal.
- j. By **1 August 2008**, the District shall submit a technical report on the overall status of compliance with implementation of BPTC and compliance with all groundwater limitations.
- k. By **1 October 2002**, the Discharger shall submit a Sanitary Sewer Overflow Response Plan (SSORP) that describes procedures for responding to sanitary sewer overflows so as to (a) minimize the volume of sewer overflow that enters surface waters, and (b) minimize the adverse effects of sewer overflows on water quality and beneficial uses.

This plan shall outline steps to be taken when an overflow or spill occurs, and provide a framework to ensure that all overflows and spills are properly identified, responded to, and reported. The SSORP shall include a posting plan, in which any posting of areas contaminated with sewage is performed at the direction of the local health services agency.

The Discharger shall maintain the SSORP in an up-to-date condition and shall amend the SSORP as necessary to accomplish the above objectives. The Discharger shall review and amend the SSORP as appropriate after each sanitary sewer overflow. The Discharger shall also ensure that the up-to-date SSORP is readily available to sewer system personnel at all times and that sewer system personnel are familiar with it.

- I. By **1 April 2003**, the Discharger shall submit a Sanitary Sewer Overflow Prevention and Maintenance Plan (SSOPMP) covering the collection systems. The SSOPMP shall describe (a) actions designed to prevent, or minimize the potential for sanitary sewer overflows, and (b) actions designed to properly maintain the collection system. The first portion of the SSOPMP shall describe how the Discharger will:

- i. Provide routine inspection and testing of all pipelines, lift stations, valves, and other key system components. The inspection/testing program shall be designed to reveal problems that might lead to accidental spills and ensure that preventive maintenance is completed;
- ii. Repair or replace old, worn out, or defective equipment;
- iii. Minimize the need for manual operation of critical systems; and
- iv. Provide spill alarms or other “fail safe” mechanisms.

The second section of the SSOPMP shall describe how the Discharger will:

- i. Properly manage, operate and maintain, at all times, all parts of the collection system;
- ii. Provide adequate capacity to convey base flows and peak flows for all parts of the collection system to the WWTP;
- iii. Take all feasible steps to stop and mitigate the impact of sanitary sewer overflows in portions of the collection system; and
- iv. Provide notification to parties with a reasonable potential for exposure to pollutants associated with the overflow event.

The SSOPMP shall contain a detailed map or maps of the sanitary sewer system, identifying sewer mains, manholes, and lift stations; a detailed listing of elements to be inspected; a description of inspection procedures and inspection frequency; and sample inspection forms.

The Discharger shall maintain the SSOPMP in an up-to-date condition and shall amend the SSOPMP whenever there is a change (e.g. in the design, construction, operation, or maintenance of the sanitary sewer system or sewer facilities) that materially affects the potential for sanitary sewer overflows, or whenever there is a sanitary sewer overflow. The Discharger shall ensure that the up-to-date SSOPP is readily available to sewer system personnel at all times and that sewer system personnel are familiar with it.

2. Upon completion of tasks set forth in Provisions E.1, the Board shall consider the evidence provided and make a determination regarding whether the Discharger has justified BPTC and the appropriate final numeric groundwater limitations that comply with Resolution 68-16.
3. If groundwater monitoring results show that the discharge of waste to the disposal field (irrigation area) is causing groundwater to contain waste constituents in concentrations statistically greater than background water quality, then within 120 days of the request of the Executive Officer, the Discharger shall submit a report showing that degradation of the groundwater complies with SWRCB Resolution No. 68-16, i.e., that it is (a) in the best interest of the people of the state, (b) that best practical treatment and control measures have been implemented to reduce the amount of degradation, (c) that the groundwater degradation will not exceed applicable water quality objectives, and (d) that the degradation is confined to a specified area.
4. The Discharger shall maintain an Operations and Maintenance (O & M) Manual which is kept current of the existing conditions and procedures. Operating personnel shall keep a copy of the O & M Manual at the wastewater treatment plant site for reference. Operating personnel shall be familiar with its contents and location, and it shall be made available for review during Regional Board inspection. At a minimum, items to be kept with the O & M Manual are as follows:
 - a. Operating procedures for plant wastewater treatment processes, irrigation, aeration, and chlorination (equipment runtimes, chlorine barrel replacement schedule) plus monitoring procedures, times, and locations;
 - b. Operation and maintenance records to include equipment maintenance schedules, operator logs, Monitoring and Reporting Program self-monitoring reports, system upsets, sludge disposal, and personnel training;
 - c. An emergency response plan to include reporting procedures, agency contact phone numbers, a spill response plan for hazardous materials, raw wastewater, partially treated wastewater, spill cleanup, and cleanup waste products disposal plan; and
 - d. A copy of this Order.

5. The Discharger shall comply with the Monitoring and Reporting Program No. 5-01-210, which is part of this Order, and any revisions thereto as ordered by the Executive Officer.
6. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."
7. The Discharger shall use the best practicable cost-effective control technique(s) currently available to comply with discharge limits specified in this order.
8. The Discharger shall not allow pollutant-free wastewater to be discharged into the wastewater collection, treatment, and disposal system in amounts that significantly diminish the system's capability to comply with this Order. Pollutant-free wastewater means rainfall, groundwater, cooling waters, and condensates that are essentially free of pollutants.
9. The Discharger shall ensure that the WWTP is operated and managed by certified wastewater treatment plant operators in accordance with Title 23 of the California Code of Regulations, Division 3, Chapter 26.
10. The Discharger shall report to the Board any toxic chemical release data it reports to the State Emergency Response Commission within 15 days of reporting the data to the Commission pursuant to section 313 of the "Emergency Planning and Community Right to Know Act of 1986."
11. The Discharger shall report promptly to the Board any material change or proposed change in the character, location, or volume of the discharge.
12. In the event of any change in control or ownership of land or waste discharge facilities presently owned or controlled by the Discharger, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be forwarded to this office.
13. The Discharger shall submit to the Board on or before each compliance report due date, the specified document or, if appropriate, a written report detailing compliance or noncompliance with the specific schedule date and task. If noncompliance is being reported, then the Discharge shall state the reasons for such noncompliance and provide an estimate of the date when the Discharger will be in compliance. The Discharger shall notify the Board in writing when it returns to compliance with the time schedule.
14. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may

result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or recession of this Order.

15. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.
16. The Board will review this Order periodically and may revise requirements when necessary.

I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 6 September 2001.

GARY M. CARLTON, Executive Officer